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Foldable bed

Abstract

A foldable bed includes a first outer panel defining a length and a width, a middle panel hingedly coupled to the first outer panel along an inner edge of the first outer panel, the middle panel defining a length and a width, the middle panel defining a cut-out having a length and a width, and a second outer panel hingedly coupled to the middle panel opposite the first outer panel and along an inner edge of the second outer panel, the second outer panel defining a length and a width. Each of the first outer panel, the middle panel, and the second outer panel includes a rigid substrate with a cushion coupled thereto, and the panels are hingedly coupled by a webbing extending across a lower surface of the foldable bed.

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Background/Summary

RELATED APPLICATIONS (1) This application claims priority to U.S. Provisional Patent Application Ser. No. 63/282,465, filed on Nov. 23, 2021, the entire disclosure of which is hereby expressly incorporated by reference.

BACKGROUND OF THE INVENTION

(1) There exists a tremendous need for drivers of day cab tractors and other types of work trucks to have a way to rest and relax during their shifts. Too often, drivers are left slumped over the steering wheel, leaning against the window or in any variation of uncomfortable positions while attempting to rest and recharge. While it may seem like a person is able to drift off to sleep in some positions, it is very frequently poor, low-quality sleep that may not recharge a driver's energy levels any more

than those levels would have been without sleep.

(2) Truck drivers, in particular, are not only required to operate on a minimum amount of sleep for safety purposes, but carry the extra burden of operating some of the largest vehicles on the road and bearing the responsibility of their own safety as well as all the vehicles, fellow drivers, and infrastructure around them. Drivers may also be out on the road at all hours of the day, and may not have access to temporary resting facilities, such as hotels. Thus, sleeping in their trucks may not only be the most optimal location to rest, but may be the only option available. In order to achieve quality rest within the confines of the truck, a bed must be properly sized with appropriate dimensions to comfortably fit a person's body, while still fitting within the truck for proper placement and positioning.

(3) Another issue that arises from use of a sleeping apparatus while out on the road is storage of the object. Storing the apparatus anywhere aside from the cabin of the day cab may not always be an option. Therefore, the apparatus may need to fit within the cab of the vehicle in order to be transported around with the driver. Developments in technology addressing each of the issues described above are therefore desirable.

SUMMARY OF THE INVENTION

(4) The apparatus described in the present disclosure is a folding bed that allows a person, e.g., the driver of the vehicle, to rest and relax while the vehicle is safely parked to recharge before setting out on the road again. The apparatus is sized to specifically fit the dimensions of a cabin of a truck, laid across the driver's and passenger's seats, while having folding capabilities for fitting within the driver's cabin for easy storage and access.

(5) In one aspect of the disclosure, a foldable bed may include a first outer panel defining a length and a width; a middle panel hingedly coupled to the first outer panel along an inner edge of the first outer panel, the middle panel defining a length and a width and defining a cut-out having a length and a width; and a second outer panel hingedly coupled to the middle panel opposite the first outer panel and along an inner edge of the second outer panel, the second outer panel defining a length and a width. Each of the first outer panel, the middle panel, and the second outer panel may include a rigid block with a cushion coupled thereto, and the panels are hingedly coupled by a webbing extending across a lower surface of the foldable bed. The middle panel may be sized and shaped to contact and rest on a driver's seat and a front passenger's seat of a vehicle and span a gap between the driver's seat and the front passenger's seat.

(6) A length of the bed may be defined by the sum of the length of the first outer panel, the length of the middle panel, and the length of the second outer panel, the length of the bed measuring between about 58 inches and about 64 inches. The width of the first outer panel may be equal to the width of the middle panel, which may be equal to the width of the second outer panel, and the width of each panel may measure between about 20 inches and about 23 inches. The length of the first outer panel and the length of the second outer panel may each measure between about 14 inches and about 15.5 inches. The length of the middle panel may measure between about 30 inches and about 33 inches. The length of the cut-out may measure between about 11 inches and about 14 inches. The width of the cut-out may measure between about 4 inches and about 6 inches.

(7) The first outer panel may be configured to fold relative to the middle panel about a hinge located between the inner edge of the first outer panel and a first edge of the middle panel to transition from a flat position to a folded position, and wherein the first outer panel includes an adhesive on a lower surface of the first outer panel configured to contact a lower surface of the middle panel to hold the first outer panel in the folded position. The second outer panel may be configured to fold relative to the middle panel about a hinge located between the inner edge of the second outer panel and a second edge of the middle panel to transition from a flat position to a folded position, and wherein the second outer panel includes an adhesive on a lower surface of the second outer panel configured to contact the lower surface of the middle panel to hold the second outer panel in the folded position. The middle panel may extend along a first plane, and when the

first outer panel and the second outer panel are in the folded positions, the first and second outer panels extend along a second plane parallel to the first plane.

(8) The foldable bed may further include an upper panel extending from an inner edge to an outer edge, the inner edge coupled to the first outer panel. The first outer panel may extend along a first plane, and the upper panel extends along a second plane oblique to the first plane. The upper panel may include at least a first support rod extending between the first plane and the second plane to hold the upper panel in an upright position. The upper panel may include an adhesive material along a portion of the inner edge contacting the first outer panel to hold the upper panel in place relative to the first outer panel. The upper panel may have a length extending between the inner edge and the outer edge, the length may measure between about 18 inches and about 21 inches. The upper panel may be configured to transition from an upright position to a flat position, wherein in the flat position, the upper panel extends along a third plane parallel to the first plane. When the upper panel is in the flat position, the upper panel may be adjacent the first outer panel, and the first outer panel may be configured to fold along a hinge between the inner edge of the first outer panel and an adjacent edge of the middle panel. The upper panel may be configured to be decoupled from the first outer panel. At least a portion of the upper panel may define a curvature.

(9) According to another aspect of the disclosure, a kit may include a foldable bed and an upper panel attachment. The foldable bed may include a first outer panel extending along a first plane; a middle panel hingedly coupled to the first outer panel along an inner edge of the first outer panel, the middle panel defining a cut-out; and a second outer panel hingedly coupled to the middle panel opposite the first outer panel and along an inner edge of the second outer panel. The upper panel attachment may be configured to be detachably coupled to the first outer panel or the middle panel such that the upper panel attachment extends along a second plane oblique to the first plane.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The foregoing features of the invention will be apparent from the following Detailed Description of the Invention, taken in connection with the accompanying drawings, in which:

(2) FIG. 1 is a top view of a foldable bed in an unfolded position according to an embodiment of the disclosure;

(3) FIG. 2 is a bottom perspective view of the foldable bed of FIG. 1;

(4) FIG. 3 is a top view of the foldable bed of FIG. 1 in a first partially folded position;

(5) FIG. 4 is a top view of the foldable bed of FIG. 1 in a second partially folded position;

(6) FIG. 5 is a top view of the foldable bed of FIG. 1 in a fully folded position;

(7) FIG. 6 is a bottom view of the foldable bed of FIG. 1 in the fully folded position;

(8) FIG. 7 is a perspective view of the foldable bed of FIG. 1 in the fully folded position;

(9) FIG. 8 is a side view of the foldable bed of FIG. 1 in the fully folded position;

(10) FIG. 9 is a perspective view of the foldable bed of FIG. 1 in an unfolded position disposed within a cabin of a vehicle;

(11) FIG. 10 is a front view of a foldable bed with an upper panel according to an embodiment of the disclosure;

(12) FIG. 11 is a front view of a foldable bed with an upper panel having a support rod according to an embodiment of the disclosure; and

(13) FIG. 12 is a top view of another foldable bed according to an embodiment of the present disclosure.

DETAILED DESCRIPTION

(14) As used herein, the terms “about,” “generally,” “approximately,” and “substantially” are intended to mean that slight deviations from absolute are included within the scope of the term so

modified.

(15) A foldable bed as described in the present disclosure is intended to be placed in the cabin of a vehicle, specifically a truck, such as all major brands of day cab tractors. Although the foldable bed described herein is intended for such use, it is not limited to the intended use. That is, the foldable bed may be sized to fit in the cabin of any vehicle, anywhere within a passenger cabin, such as across the front row including the driver's and front passenger's seats, or across a middle row of seats, a back row of seats, or in the trunk or cargo bed of a vehicle if available. Furthermore, the bed may be removed from the vehicle altogether and used as a bed anywhere outside of the vehicle.

(16) A foldable bed **100** as illustrated in FIG. **1** includes a first outer panel **110**, a middle panel **120**, and a second outer panel **130**. Each panel **110**, **120**, **130** is generally rectangular in shape. The first outer panel **110** includes an inner edge **112** and an outer edge **114**, and the second outer panel **130** includes an inner edge **132** and an outer edge **134**. The middle panel includes a first edge **122** adjacent the inner edge **112** of the first outer panel **110**, and the middle panel further includes a second edge **124** opposite the first edge **122** and adjacent the inner edge **132** of the second outer panel **130**. The first outer panel **110** is hingedly coupled to the middle panel **120** such that the first outer panel **110** is able to rotate relative to the middle panel **120**, i.e., the first outer panel **110** is able to rotate about an axis extending generally on or between the inner edge **112** of the first outer panel **110** and the first edge **122** of the middle panel **120**. The second outer panel **130** is hingedly coupled to the middle panel **120** in the same manner that the first outer panel **110** is coupled to the middle panel **120**. That is, the second outer panel **130** is hingedly coupled to the middle panel **120** such that the second outer panel **130** is able to rotate relative to the middle panel **120**, i.e., the second outer panel **130** is able to rotate about an axis extending generally on or between the second edge **124** of the middle panel **120** and the inner edge **132** of the second outer panel **130**. For example, as described in greater detail herein, a first hinge **115a** can be positioned between the first outer panel **110** and the middle panel **120** and a second hinge **115b** can be positioned between the middle panel **120** and the second outer panel **130**. The foldable bed **100** has a length defined by the distance between the outer edge **114** of the first outer panel **110** and the outer edge **134** of the second outer panel **134**. The foldable bed **100** further has a width defined in a direction perpendicular to the length from a first side edge **106** to a second side edge **108**.

(17) The middle panel **120** includes a cut-out **125** which is also generally rectangular in shape. The cut-out **125** is centered along the length of the middle panel **120** and offset along the width of the middle panel **120** such that the cut-out **125** is adjacent the first side edge **106** of the middle panel **120**.

(18) The foldable bed **100** is shown in FIG. **1** in an unfolded configuration or an unfolded position. In the unfolded position, each of the panels **110**, **120**, **130** extends along a single plane. In other words, the middle panel **120** extends along a first plane and, when the first and second outer panels **110**, **130** are in the unfolded position, the outer panels also extend along the first plane. In the unfolded position, the foldable bed **100** is configured to be positioned in the cabin of a vehicle, such as that of a cargo truck, over the driver's and passenger's seats. For example, as shown in FIG. **9**, at least a portion of the first outer panel **110** may be placed on top of the driver's seat **151** and at least a portion of the second outer panel **130** may be placed on top of the passenger's seat **152** in the same row as the driver's seat **151**. It is also contemplated that at least a portion of the middle panel **120** may be placed on top of the driver's seat and the passenger's seat (e.g., a portion of the middle panel **120** proximate the first outer panel **110** may be disposed on the driver's seat **151** and a portion of the middle panel **120** proximate the second outer panel **130** may be disposed on the passenger's seat **152**). In some examples, at least a portion of the middle panel **120** may be suspended between the driver's seat **151** and the passenger's seat **152**. In other examples, the truck may include seating or support extending across the entire row from the driver's seat **151** to the passenger's seat **152**, and the foldable bed **100** may be placed thereon.

(19) The cut-out **125** is carved out of the middle panel **120** so that the middle panel **120** is generally

contoured around potentially interfering elements such as a gear shifter **153**, as also shown in FIG. 9. As such, the gear shifter **153** may pass through the cut-out **125** when the foldable bed **100** is in the unfolded position placed on top of the driver's seat **151** and passenger seat **152** so that the foldable bed **100** may lay substantially flat on the seats without any protrusions or interruptions. The foldable bed **100** may lay across the width of the vehicle, approximately from the interior of a driver's side door (or proximate thereto) to the interior of a passenger's side door (or proximate thereto) when the doors of the cabin are closed. Such positioning allows a person, such as the driver, to comfortably lay across the width of the vehicle (i.e., along the length of the bed) to rest while the vehicle is parked and stationary.

(20) Although the foldable bed **100** may have any dimensions suitable for fitting the bed within any sized vehicle, the foldable bed **100** illustrated in FIGS. 1-12 has particular dimensions for fitting within the cabin of a particularly sized truck, such as all major brands of day cab tractors. The foldable bed **100** has a length extending from the outer edge **114** of the first outer panel **110** to the outer edge **134** of the second outer panel **130** measuring between about 58 inches and about 64 inches, preferably about 61 inches. The first outer panel **110** has a length extending from outer edge **114** to inner edge **112** measuring between about 14 inches and about 15.5 inches, preferably about 14.75 inches. The middle panel **120** has a length extending from the first edge **122** of the middle panel **120** to the second edge **124** of the middle panel **120** measuring between about 30 inches and about 33 inches, preferably about 31.5 inches. The length of the middle panel **120** is preferably sufficient to span length of a gap between a driver's seat and a passenger's seat such that the middle panel **120** at least contacts both seats and a load applied to the middle panel **120** can be distributed between both seats. The second outer panel **130** has a length extending from the inner edge **132** of the second outer panel **130** to the outer edge **134** of the second outer panel **130** measuring between about 14 inches and about 15.5 inches, preferably about 14.75 inches.

(21) The foldable bed **100** has a width extending from a first side edge **106** to a second side edge **108** measuring between about 20 inches and about 23 inches, preferably about 21.5 inches. Each of the panels **110**, **120**, **130** has approximately the same width. The cut-out **125** is carved out of the first side edge **106** of the middle panel **120** along approximately the center of the length of the middle panel **120** and the foldable bed **100** such that the foldable bed **100** is symmetrical about a center axis extending across the width of the bed **100**. The cut-out **125** has a length measuring between about 11 inches and about 14 inches, preferably about 12.5 inches. The cut-out **125** has a width measuring between about 4 inches and about 6 inches, preferably about 5 inches. The particular dimensions of the foldable bed **100** are critical for fitting within the confines of the desired vehicle to ensure the driver is provided with effective and sufficient rest to be able to resume driving. Although not limited to the disclosed dimensions, the precise size disclosed herein optimizes the surface area of the foldable bed **100** which can be used within the vehicle to comfortably support a person's weight, while simultaneously remaining capable of being stored neatly and out of the way.

(22) As shown in FIG. 8, each panel **110**, **120**, **130** can be formed of a rigid block, substrate, panel or the like (hereinafter, substrate **107**), such as panel of wood, plastic, or other suitable material, sized accordingly for the respective panel. As shown in FIG. 1, according to some embodiments of the present disclosure, one or more of the panels **110**, **120**, **130** of the present disclosure can each include a respective rigid support member **119a-c** (together, support members **119**) configured to provide enhanced structural rigidity where the panels are required to support an enhanced load (e.g., a heavier driver). The rigid support members **119** can be formed from metal (e.g., $\frac{1}{8}'' \times \frac{1}{4}''$ extruded steel stock), or any other similarly rigid structural members. The substrate **107** for each panel **110**, **120**, **130** may include a cushion or cushioning material **109** (such as foam or other suitable material) applied to an upper surface thereof and a cover **111** (such as nylon or other suitable material) sewn, stapled, or otherwise attached about the substrate **107** and cushion **109** to hold the cushion **109** to the substrate **107**. The cushion **109** may have any variation or any degree

of firmness. The thickness of the cushion **109** may range between about one inch to about two inches to provide sufficient support for a person to lay comfortably on the foldable bed **100**, while still maintaining the ability to fold the bed **100** and keep it space-efficient. Each of the panels **110**, **120**, **130** can be held together by a webbing **113** spanning across the lower surfaces **104a-c** of each of the panels **110**, **120**, **130** and coupled thereto (e.g., by sewing, stapling, or other suitable means). The webbing **113** can be formed of nylon, plastic or other strong and flexible material suitable for holding the panels **110**, **120**, **130** together and forming hinges **115a**, **115b** such that the first and second outer panels **110**, **130** can rotate with respect to the middle panel **120** as described above. According to some embodiments of the present disclosure, piano hinges, or other suitable mechanical hinge-type devices, can be positioned between adjacent panels **110**, **120**, **130** affixed (e.g., by, nails, screws, staples, adhesives, or other suitable means) thereto, thereby coupling the panels **110**, **120**, **130** together and providing for rotational movement thereabout. According to additional embodiments of the present disclosure, one or more of the substrates **107** of the panels **110**, **120**, **130** can be integrally formed (e.g., through an injection molding process or the like) from a plastic, or other suitably rigid and flexible material. According to such embodiments, the hinges **115a**, **115b** can be living hinges formed between adjacent panels **110**, **120**, **130**, providing for rotational movement thereabout, as described above.

(23) The components from which the foldable bed **100** is constructed (e.g., a rigid substrate **107** with a covered cushion **109** and webbing **113** coupling each of the rigid panels **110**, **120**, **130** to one another) provide unique advantages to the foldable bed **100**. For instance, the rigid panels **110**, **120**, **130** provide support to bear load while maintaining a minimal thickness of the foldable bed **100**. This allows the bed **100** to be folded with the flat rigid panels **110**, **120**, **130** abutting each other, and the minimized thickness of the panels reduces the total thickness of the bed **100** in the folded position relative to a bed formed entirely of cushion. The cushion **109** of the present invention is sufficiently thick to optimize comfort for the user while also optimizing the thickness of the bed **100** in the folded position for storage. Further, the webbing **113** extending across the lower surface **104** of the bed **100** is a strong, lightweight material able to hold the panels together and hinge with respect to each other without interference, and with a minimal risk of malfunction.

(24) According to some embodiments of the present disclosure, one or more secondary panels **117a-c** (together, secondary panels **117**) can be affixed to the webbing **113** on the underside of the foldable bed **100** (see FIG. 2 and FIG. 8), thereby sandwiching the webbing **113** between the substrates **107** and the secondary panels **117**. The secondary panels **117**, in combination with the webbing **113**, function to providing additional structural rigidity to the hinges **115** of the foldable bed **100**, thereby preventing the panels **110**, **120**, **130** of the foldable bed **100** from twisting about a longitudinal axis of the foldable bed **100** or pivoting in an unwanted direction about the hinges **115**. The secondary panels **117** can be formed from a thermoplastic, or another similarly durable and resilient material, and the webbing **113** can be formed of a flexible polypropylene material, such as Propex Curv® sold and distributed by Rockywoods Fabrics LLC, or another similarly flexible and resilient material. It is contemplated that the panels **110**, **120**, **130** may be any shape, such as square, trapezoidal, rhombus, etc. The panels **110**, **120**, **130** may be shaped and sized to fit within any vehicle. It is contemplated that the cut-out **125** may be any shape (e.g., semi-circular, triangular, square, trapezoidal, etc.) and may be located anywhere along the middle panel, including along an edge or between the edges. A cut-out may also be included on either or both of the first and second outer panels **110**, **130** to accommodate any other potential elements from interfering with the foldable bed **100** from lying flat across the seats.

(25) The foldable bed **100** may be transitioned from the unfolded position to a folded configuration or position. The first outer panel **110** may be rotated about its hinge axis such that a lower surface **104a** of the first outer panel **110** is rotated toward a lower surface **104b** of the middle panel **120** until the lower surfaces **104a**, **104b** abut one another, as shown in FIG. 3, which illustrates the foldable bed **100** in a partially folded position. The lower surfaces **104a**, **104b** of the first outer

panel **110** and the middle panel **120** may each include corresponding adhesives or fixation devices **105a**, **105b**, such as a hook and loop material, magnets, or the like, to detachably couple the first outer panel **110** and the middle panel **120** by their respective lower surfaces **104a**, **104b** and hold the foldable bed **100** in the folded position. Further, the second outer panel **130** may be rotated about its hinge axis such that a lower surface **104c** of the second outer panel **130** is rotated toward the lower surface **104b** of the middle panel **120** until the lower surfaces **104b**, **104c** abut one another, as shown in FIG. 4, which illustrates the foldable bed **100** in a partially folded position. The respective lower surfaces **104c**, **104b** of the second outer panel **130** and the middle panel **120** may each include corresponding adhesives or fixation devices **105c**, **105b**, such as a hook and loop material, magnets, or the like, to detachably couple the second outer panel **130** and the middle panel **120** by their respective lower surfaces **104c**, **104b** and hold the foldable bed **100** in the folded position.

(26) The foldable bed **100** is in the folded position when the lower surfaces **104a**, **104c** of both the first and second outer panels **110**, **130** abut the lower surface **104b** of the middle panel **120**, as shown in FIGS. 5-6. In the folded position, the first and second outer panels **110**, **130** no longer extend along the first plane (i.e., the plane along which the middle panel extends), but instead the outer panels **110**, **130** extend along a same second plane which is parallel to the first plane. In the folded position, the length of the foldable bed **100** is reduced so that the bed **100** is more compact and takes up less space in the longitudinal direction. The folded position optimizes the size of the foldable bed **100** so that it can be placed and stored in a convenient location without being a nuisance or interference while the bed **100** is not in use. For example, a driver can fit the foldable bed **100** in the folded position behind the passenger's seat or behind the driver's seat within the cabin. The ability to store the foldable bed **100** in a convenient location such as behind one or both of the seats allows for easy access to the bed **100** to deploy onto the seats or stow the bed away without hassle. It is also contemplated that the first and second outer panels **110**, **130** may be configured to rotate in the opposite direction such that upper surfaces **102a**, **102c** of the first and second outer panels **110**, **130** abut an upper surface **102b** of the middle panel **120** in the folded position. However, in some examples, as in the example described above, the nylon webbing **113**, or other hinge mechanism (e.g., hinges **115a**, **115b**), holding the panels **110**, **120**, **130** together may only allow the first and second outer panels **110**, **130** to rotate in the direction in which their lower surfaces **104** face, and may prevent the panels from rotating in the opposite direction beyond the flattened (unfolded) configuration of the foldable bed **100** in which each of the panels **110**, **120**, **130** lies in the same first plane.

(27) As shown in FIGS. 10 and 11, the foldable bed **100** may optionally include an upper panel **140** extending from an inner edge **142** to an outer edge **144**. The upper panel **140** may be formed in substantially the same manner as the other panels **110**, **120**, **130**, e.g., including a rigid block, substrate, panel or the like, such as wood or plastic, with a cushioning applied thereon, such as foam, and a cover sewn or otherwise attached to the rigid panel over the cushion to maintain the placement of the cushion (see, e.g., FIG. 8). The upper panel **140** has a length extending between the inner edge **142** and the outer edge **144** measuring between about 18 inches and about 21 inches, preferably about 19.5 inches. The upper panel **140** may be coupled to the foldable bed **100** along the inner edge **142** of the upper panel **140**, and the inner edge **142** may be coupled anywhere along the bed, e.g., the first outer panel **110**. The upper panel **140** is shown in an upright position in FIG. 10. In the upright position, the upper panel **140** extends along a third plane, the third plane being oblique to and intersecting the first plane along which the panels **110**, **120**, **130** extend in the unfolded position. In some examples, such as the example shown in FIG. 10, the upper panel **140** may be detachably coupled to the first outer panel **110** at the inner edge **142** of the upper panel **140** and the outer edge **114** of the first outer panel **110**. The upper panel **140** may be attached to the first outer panel **110** by an adhesive or fixation device of any kind, such as a hook and loop material, and the upper panel **140** may then be placed in an upright position as shown in FIG. 10 to lean

against the interior of the driver's side door when the door is closed.

(28) The upper panel **140** may alternatively be coupled to the outer edge **134** of the second outer panel **130** and lean against the interior of the passenger's side door when the door is closed. One or more of the upper panel **140** may be included in a kit or as an additional attachment piece either together with or separate from the foldable bed **100**, and the upper panel **140** may extend from the foldable bed forming an angle anywhere between 0 and 90 degrees relative to the first plane along which the panels of the bed extend in the unfolded position. The upright positioning can improve comfort for a user and thereby improve quality of rest by providing an inclined surface for a person to lay at a more natural angle rather than completely flat across a single plane, with the interior of the door bearing the weight of the person laying on the foldable bed **100**. It is also contemplated that the upper panel **100** may have any degree of curvature to further improve the positioning angle or cater to a user's preferred sleeping position.

(29) In other examples, the upper panel **140** may be detachably coupled to any one of the panels **110**, **120**, **130** or a portion therebetween. For example, as shown in FIG. **11**, the inner edge **142** of the upper panel **140** may be detachably coupled at the hinge or axis (e.g., at edge **112**) between the first outer panel **110** and the middle panel **120**. The inner edge **142** of the upper panel **140** may include an adhesive or fixation device applied thereon, such as a hook and loop material, that can be used to place the upper panel **140** in a desired position to detachably fix the inner edge **142** of the upper panel **140** relative to the foldable bed **100**. The upper panel **140** may then be held in the upright position by a support rod **145** as described below in greater detail. When use of the foldable bed **100** with the upright panel **140** attachment is finished, the upper panel **140** may be decoupled from the foldable bed **100** and stored away in a location such as those described above, wherein in the detached upper panel **140** may be placed adjacent the foldable bed **100** in the folded position such that the upper panel **140** lies in a plane extending parallel to the planes of the panels of the foldable bed **100**. Alternatively, because the upper panel **140** is detachable, it may be stored completely separately from the foldable bed **100** to improve space efficiency of the vehicle in which the foldable bed **100** is used in.

(30) In some examples, the upper panel **140** may be hingedly coupled to any one of the first outer panel **110**, the middle panel **120**, or the second outer panel **130** or anywhere therebetween. For example, the inner edge **142** of the upper panel **140** may be hingedly coupled at the hinge or axis between the first outer panel **110** and the middle panel **120**. In such examples, the upper panel **140** may be fixed to the other panels and adapted to fold with the bed as the foldable bed **100** transitions to the folded position. That is, when the first outer panel **110** is in the folded position as described above with the lower surface **104** of the first outer panel **110** abutting the lower surface **104** of the middle panel **120**, the upper panel **140** may also rotate about its hinge and fold in a first direction such that the lower surface **148** of the upper panel **140** follows the first outer panel **110**, and the lower surface **148** of the upper panel **140** abuts the upper surface **102** of the first outer panel **110** in the folded position. Alternatively, the upper panel **140** may rotate about its hinge in a second direction opposite the first direction, such that an upper surface **146a** of the upper panel **140** is rotated toward the upper surface **102b** of the middle panel **120** and the surfaces abut one another in the folded position. In either manner of folding described above, the length of the foldable bed **100** is reduced so that the bed is more compact and takes up less space in the longitudinal direction. The folded position optimizes the size of the foldable bed **100** so that it can be placed and stored in a convenient location without being a nuisance or interference while the bed is not in use. For example, a driver can fit the foldable bed **100** in the folded position behind the passenger's seat or behind the driver's seat within the cabin. The ability to store the foldable bed **100** in a convenient location such as behind one or both of the seats allows for easy access to the bed to deploy or stow the bed away without hassle.

(31) In some examples, the upper panel **140** may include a means for remaining in the upright position, particularly when a load is applied to the upper panel **140**, such as the weight of at least a

portion of a person's body, and in examples in which the upper panel **140** is not positioned to lean against the interior of the door of the vehicle. For example, the foldable bed **100** may include at least one support rod **145** extending between a lower surface **146b** of the upper panel **140** and the upper surface **102a** of the first outer panel **110**, as shown in FIG. **11**. In some examples, the foldable bed **100** may include a single support rod **145** positioned to extend from the first outer panel **110** generally in the middle along the length and width of the first outer panel **110**. Alternatively, the foldable bed **100** may include a plurality of support rods **145** extending from one of the side edges **106**, **108** of the first outer panel **110** or the outer edge **114** of the first outer panel **110** up to a surface or edge of the upper panel **140**. The support rod **145** may extend perpendicular to the first plane (i.e., the plane along which the first outer panel **110** extends) or oblique to the first plane, e.g., perpendicular to the third plane (i.e., the plane along which the upper panel **140** extends).

(32) At least one support rod **145** may be detachably coupleable to the upper panel **140** to hold the panel in the upright position. Alternatively, at least one support rod **145** may be hingedly coupled to the lower surface **146b** of the upper panel **140** such that the support rod **145** can be adjusted to transition from a protruding position to a folded position. That is, in the folded position, the support rod **145** may have a length which extends generally parallel to plane along which the upper panel **140** extends, and the support rod **145** may abut the lower surface **146** of the upper panel **140**, with a first end of the support rod **145** hingedly coupled to an edge or the lower surface **146b** of the upper panel **140**. Any portion of the support rod **145**, such as a second end opposite a first end, may be pulled to separate the support rod **145** from the lower surface **146** of the upper panel **140** such that the support rod **145** rotates about its hinge. The support rod **145** may rotate anywhere between 0 and 180 degrees, e.g., approximately 90 degrees so that its length extends along an axis oblique to the plane along which the upper panel **140** extends, and thus the second end of the support rod **145** is able to contact a structure below, such as the first outer panel **110**, to hold the upper panel **140** in the upright position. When it is then desired to transition the upper panel **140** from the upright position to the folded position (or remove the upper panel **140** altogether), the upper panel **140** may simply be lifted to remove the load applied to the support rod **145**, and the support rod **145** may thereafter be folded in the opposite direction it was unfolded to revert the support rod to its folded position flush against the lower surface **146b** of the upper panel **140**.

(33) FIG. **12** is a top view of another foldable bed according to present disclosure (hereinafter, foldable bed **200**). Foldable bed **200** can be substantially similar in both form and function to foldable bed **100**, discussed in connection with FIGS. **1-11**, except for distinctions noted herein. Accordingly, foldable bed **200** can include a first outer panel **210** coupled to a middle panel **220** by way of a hinge **215a** and a second outer panel **230** coupled to the middle panel **220** by way of a hinge **215b**. An exterior periphery of the foldable bed **200** is defined by a right edge **214**, a left edge **234**, a front edge **206**, a rear edge **208**, and a cut-out **225** disposed on the front edge **206** of the middle panel **220**. The foldable bed **200** can also include a power supply **250** (e.g., a 12V DC male power connector for receiving electrical power from a 12V DC female power socket of a vehicle and/or a battery, such as a rechargeable battery pack) for providing electrical power to one or more electrical components of the foldable bed **200**.

(34) According to one example, the power supply **250** can provide power to one or more sensors **252** disposed within one or more of the panels **210**, **220**, **230** for determining when the foldable bed **200** is bearing weight (e.g., determining when a user is resting on the foldable bed **200**). As will be understood by those of ordinary skill in the art, the sensor **252** can be a weight sensor, pressure sensor, proximity sensor, switch, or any other means configured to determine when a user is resting on the foldable bed **200**. The sensor **252** can also include a communications interface (e.g., serial interface, Bluetooth, Wi-Fi, cellular, etc.) for transmitting sensor data to another device, such as, but not limited to, a memory disposed within the foldable bed **200**, a driver's mobile device, an electronic log device (ELD) associated with the driver and/or vehicle, a computer system or server

of a trucking company, a computer system or server controlled by the Federal Motor Carrier Safety Administration (FMSCA) of the United States Department of Transportation (USDOT), or any other computing device. Accordingly, the sensor **252** can record data that can be stored and later analyzed to determine periods of time when the driver of the vehicle is resting on the foldable bed **200**. Furthermore, this sensor data can be accessed by the FMSCA during an audit to determine if a driver and/or trucking company is complying with federally mandated guidelines related to maximum driving times. Further still, by using the sensor **252** to record periods of rest, drivers can “split their time” between periods of rest to increase the maximum amount of driving time in a given week by up to twenty-one (21) hours.

(35) According to another example, the power supply **250** can provide power to a heating element **254**, which can be disposed within one or more of the panels **210**, **220**, **230** and can be adjustably activated to heat the foldable bed **200** when desired. Similarly, the power supply **250** can provide power to one or more massaging elements or features **256**, which can be disposed within one or more of the panels **210**, **220**, **230** and can be adjustably activated to provide a massaging action when desired. The electrical components of the foldable bed **200** can also include, but are not limited to, one or more of a cigarette lighter, a heated or cooled cup holder, a reading light and one or more USB ports, a wireless charging pad, an AC power inverter, or other means for providing power to another electronic device (e.g., mobile device, laptop, gaming device, or the like). The foldable bed **200** can also include one or more of a keychain holder and a storage bag, tray, caddy, or lockbox configured to hold and/or organize one or more of a driver's wallet, sunglasses, pens, and other personal items.

(36) According to some embodiments of the present disclosure, the foldable bed **200** can include a modular connection interface **258** for coupling one or more modular attachments **260** to the foldable bed **200**. The modular connection interface **258** can be disposed about the periphery of the foldable bed **200**, defined by the right edge **214**, the left edge **234**, the front edge **206**, the rear edge **208**, and the cut-out **225**, and includes a coupling mechanism for attaching one or more of the modular attachments **260** to the foldable bed **200**, for example, by way of a reciprocal coupling mechanisms disposed on the modular attachments **260**. The modular connection interface **258** can also include an electrical communication interface coupled to the power supply **250** and configured to provide power to one or more of the modular attachments **260**, for example, by way of reciprocal electrical communication interfaces of the modular attachments **260**. Accordingly, the modular attachments **260** can be mechanically and electrically coupled to the foldable bed **200** and can include one or more of the cigarette lighter, the heated or cooled cup holder, the reading light, the USB ports, the wireless charging pad, the AC power inverter, the keychain holder, and the storage bag, tray, caddy, or lockbox described above.

(37) Although the invention herein has been described with reference to particular embodiments, it is to be understood that these embodiments are merely illustrative of the principles and applications of the present invention. It is therefore to be understood that numerous modifications may be made to the illustrative embodiments and that other arrangements may be devised without departing from the spirit and scope of the present invention as defined by the appended claims.

Claims

1. A foldable bed comprising: a first outer panel defining a length and a width; a middle panel hingedly coupled to the first outer panel along a first inner edge of the first outer panel, the middle panel defining a length and a width, the middle panel defining a cut-out having a length and a width; a second outer panel hingedly coupled to the middle panel opposite the first outer panel and along a second inner edge of the second outer panel, the second outer panel defining a length and a width; an upper panel extending from the first inner edge or the second inner edge; wherein each of the first outer panel, the middle panel, the upper panel, and the second outer panel includes a rigid

block with a cushion coupled thereto; and wherein each of the first outer panel, the middle panel, and the second outer panel are hingedly coupled by a webbing extending across a lower surface of the foldable bed.

2. The foldable bed of claim 1, wherein the middle panel is sized and shaped to contact and rest on a driver's seat and a front passenger's seat of a vehicle and span a gap between the driver's seat and the front passenger's seat.

3. The foldable bed of claim 1, wherein a length of the bed is defined by the sum of the length of the first outer panel, the length of the middle panel, and the length of the second outer panel, the length of the bed measuring between about 58 inches and about 64 inches.

4. The foldable bed of claim 1, wherein the length of the cut-out measures between about 11 inches and about 14 inches.

5. The foldable bed of claim 1, wherein the first outer panel is configured to fold relative to the middle panel about a hinge located between the first inner edge of the first outer panel and a first edge of the middle panel to transition from a flat position to a folded position, and wherein the first outer panel includes a first fixation device on a lower surface of the first outer panel configured to contact a reciprocal second fixation device on a lower surface of the middle panel to hold the first outer panel in the folded position.

6. The foldable bed of claim 5, wherein the second outer panel is configured to fold relative to the middle panel about a hinge located between the second inner edge of the second outer panel and the second edge of the middle panel to transition from the flat position to a folded position, and wherein the second outer panel includes a third fixation device on a lower surface of the second outer panel configured to contact the second reciprocal fixation device on the lower surface of the middle panel to hold the second outer panel in the folded position.

7. The foldable bed of claim 6, wherein the middle panel extends along a first plane, and when the first outer panel and the second outer panel are in the folded position, the first and second outer panels extend along a second plane parallel to the first plane.

8. The foldable bed of claim 1, wherein: the the upper panel is removably coupled to the first outer panel along the first inner edge, wherein the first outer panel extends along a first plane, and the upper panel extends along a second plane oblique to the first plane.

9. The foldable bed of claim 8, further comprising one or more support rods disposed between the upper panel and the first outer panel to hold the upper panel in the second plane.

10. The foldable bed of claim 8, wherein the upper panel is configured to transition from an upright position to a flat position, wherein in the flat position, the upper panel extends along a third plane parallel to the first plane.

11. The foldable bed of claim 8, wherein at least a portion of the upper panel defines a curvature.

12. A foldable bed comprising: a first outer panel defining a length and a width; a middle panel hingedly coupled to the first outer panel along a first inner edge of the first outer panel defining a first living hinge between the first outer panel and the middle panel, the middle panel defining a length and a width, the middle panel defining a cut-out having a length and a width; a second outer panel hingedly coupled to the middle panel opposite the first outer panel and along a second inner edge of the second outer panel defining a second living hinge between the second outer panel and the middle panel, the second outer panel defining a length and a width; an upper panel extending from the first living hinge or the second living hinge; wherein each of the first outer panel, the middle panel, the upper panel, and the second outer panel includes a rigid substrate and a cushion coupled to the rigid substrate; and wherein the rigid substrates of the first outer panel, the middle panel, and the second outer panel are integrally formed, the rigid substrates of the first outer panel and the middle panel being hingedly coupled by way of the first living hinge along the first inner edge, the rigid substrates of the middle panel and the second outer panel being hingedly coupled by way of the second living hinge along the second inner edge.

13. The foldable bed of claim 12, wherein the first outer panel is configured to fold relative to the

middle panel about the first living hinge, located between the first inner edge of the first outer panel and a first edge of the middle panel, to transition from a flat position to a folded position, and wherein the first outer panel includes a first fixation device on a lower surface of the first outer panel configured to contact a reciprocal second fixation device on a lower surface of the middle panel to hold the first outer panel in the folded position.

14. The foldable bed of claim 13, wherein the second outer panel is configured to fold relative to the middle panel about the second living hinge, located between the inner edge of the second outer panel and the second edge of the middle panel, to transition from the flat position to the folded position, and wherein the second outer panel includes the third fixation device on a lower surface of the second outer panel configured to contact the second reciprocal fixation device on the lower surface of the middle panel to hold the second outer panel in the folded position.

15. The foldable bed of claim 14, wherein the first living hinge is configured such that the first outer panel is prevented from rotating about the first living hinge beyond a position where the first outer panel and the middle panel are in an extended coplanar configuration.

16. The foldable bed of claim 15, wherein the second living hinge is configured such that the second outer panel is prevented from rotating about the second living hinge beyond a position where the second outer panel and the middle panel are in an extended coplanar configuration.

17. The foldable bed of claim 12, wherein a the upper panel is removably coupled to the first outer panel along the first living hinge or the second living hinge, wherein the first outer panel extends along a first plane, and the upper panel extends along a second plane oblique to the first plane.

18. The foldable bed of claim 12, further comprising one or more support rods disposed between the upper panel and the first outer panel to hold the upper panel in the second plane.

19. The foldable bed of claim 17, wherein the upper panel is positionable in an upright position where the upper panel extends along the second plane oblique to the first plane and a flat position where the upper panel extends along a third plane parallel to the first plane.

20. A kit comprising: a foldable bed comprising: a first outer panel extending along a first plane; a middle panel hingedly coupled to the first outer panel along a first inner edge of the first outer panel, the middle panel defining a cut-out; a second outer panel hingedly coupled to the middle panel opposite the first outer panel and along a second inner edge of the second outer panel; and an upper panel attachment configured to be detachably coupled to the foldable bed along the first inner edge or the second inner edge such that the upper panel attachment is positionable in an upright position where the upper panel attachment extends along a second plane oblique to the first plane and a flat position where the upper panel attachment extends along a third plane parallel to the first plane.
